

Single Answer Multiple Choice Questions

Q1. **Answer:** D

Note: This exam is labeled as Set D.

Q2. **Answer:** C

Note: A negative cosine similarity for standard non-zero TF-IDF vectors suggests an implementation issue, since cosine similarity for such vectors should be in $[0, 1]$.

Q3. **Answer:** A

Note: Proper initialization keeps gradients at a reasonable scale across layers.

Q4. **Answer:** D

Note: A composition of linear layers without nonlinearities is equivalent to a single linear transformation.

Q5. **Answer:** C

Note: Vanilla RNNs suffer from vanishing and exploding gradients over long time spans.

Q6. **Answer:** B

Note: The Markov assumption restricts the conditioning context, reducing the number of distinct n -gram contexts.

Q7. **Answer:** D

Note: Step activations provide no useful local gradient signal for gradient descent almost everywhere.

Q8. **Answer:** B

Note: Distributed word embeddings allow compositional generalization to unseen but related phrases.

Q9. **Answer:** A

Note: A bag-of-words representation with TF-IDF relies on exact lexical overlap and ignores synonymy.

Q10. **Answer:** C

Note: Higher capacity improves training fit but can hurt test performance due to overfitting.

Q11. **Answer:** D

Note: In RNNs, the same weights are reused at each time step and the loss at time t depends on the hidden state from $t - 1$.

- Q12. **Answer:** B
Note: L2 regularization penalizes large weights, encouraging smaller magnitudes that improve generalization.
- Q13. **Answer:** B
Note: When a sigmoid saturates, its derivative is near zero, leading to very small gradients.
- Q14. **Answer:** D
Note: Sharing parameters across time reduces the number of parameters and applies the same transformation at each step.
- Q15. **Answer:** A
Note: RNN language models typically use cross-entropy loss over the vocabulary.
- Q16. **Answer:** B
Note: Treating whole sentences as tokens yields very short sequences but an extremely large vocabulary.
- Q17. **Answer:** B
Note: Unrolling builds a time-indexed computation graph so gradients can flow through time during backpropagation.
- Q18. **Answer:** B
Note: LSTMs use gated mechanisms to control information flow and alleviate basic RNN gradient issues.
- Q19. **Answer:** A
Note: A BiLSTM is a standard architecture for sequence labeling tasks like NER.
- Q20. **Answer:** D
Note: If all first-layer pre-activations are negative, ReLU units output zero and the network produces all zeros.
- Q21. **Answer:** C
Note: Zero weights with non-zero biases give the same pre-activation and thus the same output for every input.
- Q22. **Answer:** C
Note: SGD uses unbiased but noisy gradient estimates, resulting in higher variance than full-batch updates.
- Q23. **Answer:** C
Note: BPE breaks rare words into subword units, allowing representation of many previously OOV forms.
- Q24. **Answer:** B
Note: With teacher forcing, the model is trained on gold histories but at test time must condition on its own predictions, causing exposure bias.

- Q25. **Answer:** C
Note: In LSTMs, the cell state c is designed to carry long-term summary information.
- Q26. **Answer:** D
Note: With non-negative TF-IDF vectors, cosine similarity 0 implies no shared non-zero features (no shared terms).
- Q27. **Answer:** A
Note: Token IDs allow the model to consume text as numerical input via a fixed vocabulary mapping.
- Q28. **Answer:** B
Note: $P(H) \cdot P(H \rightarrow C) \cdot P(C \rightarrow H) = 0.5 \times 0.3 \times 0.4 = 0.06$.
- Q29. **Answer:** D
Note: Perplexity measures how well a model fits sequence likelihood, which need not correlate perfectly with downstream task performance.
- Q30. **Answer:** D
Note: A learning rate that is too large causes updates to overshoot minima, leading to oscillating loss.
- Q31. **Answer:** D
Note: The sequence probability is $0.5 \times 0.4 \times 0.2 \times 0.1 = 0.004$ and perplexity $= 0.004^{-1/4} \approx 3.976$.
- Q32. **Answer:** A
Note: This finite Markov chain is irreducible and aperiodic, so state probabilities converge to a unique stationary distribution independent of the start state.
- Q33. **Answer:** B
Note: Skip-gram with negative sampling maximizes $\log \sigma(v_c^\top v_w) + \sum_{i=1}^k \log \sigma(-v_{n_i}^\top v_w)$.
- Q34. **Answer:** C
Note: ELMo produces context-dependent word representations, whereas Word2Vec assigns a single vector per word type.

End of Answer Key